

XAU/USD System Rebuild — Build Roadmap & Requirements

Companion to: Master Context v2.0 + Audit (P0–P13) Status as of: June 2, 2026 — build starts tomorrow

1. What we're building (confirm this first)

A **discretionary** trading system for the FundingPips challenge, supported by a research/decision lab. **Nothing is automated against the broker.** You make every trade by hand off the chart; the lab tells you where your edge is and keeps you inside your risk envelope.

The design rests on one principle: **a hard line between what is frozen and what is fluid.**

- **Frozen** (set once, then left alone): risk rails, the measurement/journaling schema, the data pipeline, the event calendar. These protect the account and make every experiment comparable.
- **Fluid** (change as often as you want): factors, thresholds, regime definitions, setup variants. Because the frozen layer never moves, you can rip out and replace any signal on any day and the comparison stays apples-to-apples.

That separation is the "never stop exploring" guarantee.

The three surfaces

Surface	Lives in	Role	You...
1 — Execution	TradingView / Pine (LSE v5 HUD)	Live session, IB, VWAP, structure	Act (manual, human)
2 — The Lab	Python, offline	Feature engine + regime model + expectancy harness	Explore (test edge on history)
3 — The Daily Brief	Python output, read alongside the HUD	Scored bias + risk envelope + gating flags	Decide (informed, still your call)

Two disciplines baked in from line one

1. **No lookahead** — the lab uses confirmed bar values only (your HTF bias and pseudo-POC both repaint intrabar, so naive use would inflate results).
 2. **Costs always on** — every backtested trade pays modeled spread + slippage *before* its R is counted (P12). A backtest without this feels great and lies.
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2. The build roadmap

Ordered so the **first brick produces a visible payoff**, not just plumbing. Each phase ends when its deliverable exists and you've approved it.

Phase 0 — Foundation (*mostly done* — see §3)

Data pipeline + architecture decision + existing Pine HUD. The runway.

Phase 1 — The Lab + P0 (Regime) + first expectancy table ← **the magic moment**

- Stand up the Python research repo (loaders, structure).
- **Feature engine**: reconstruct the Pine features in Python from M5 history, confirmed-bar only.
- **P0 — Regime classifier**: fit a Range/Trend structural regime (Markov) on D1. Keep the existing ATR-ratio as a *separate* volatility axis (two axes, not one).
- **Expectancy harness v1**: replay your setups (sweep S2/S3, drift, Tokyo break) across history, sliced by **setup × session × regime** → hit-rate, avg-R, profit factor, sample size.
- Mirror the regime label back into the Pine HUD so chart and lab agree.
- **Deliverable**: the first table showing *where your edge actually lives*.

Phase 2 — Account defense (P2) — freeze the risk rails

- Fixed-fractional risk (0.25–0.5% per trade).
- Hard daily-loss circuit breaker set *below* the FundingPips limit.
- Correlated-exposure cap (gold + silver + short-USD = one bet).
- Fractional-Kelly sizer — *parked* until Phase 1 gives a real edge estimate (Kelly is undefined without one).
- **Deliverable**: a frozen risk envelope feeding the Daily Brief.

Phase 3 — Orthogonalized scoring (P1) + thresholds (P3) + expectancy formalized (P4)

- **P1 — PCA:** collapse the dollar/rates cluster (DXY-proxy, 10y nominal, 10y real, EUR/USD, USD/JPY) into one composite, and the price-structure cluster (HTF bias, IB state, imbalance, POC, VWAP) into another. Score drivers, not symptoms.
- **P3:** define the real max score; symmetrize long/short thresholds (fix the current asymmetry).
- **P4:** lock the journaling/tagging schema so the **live journal and the backtest use identical tags** — forward results become directly comparable to historical ones.
- **Deliverable:** the Daily Brief takes its real shape.

Phase 4 — Signal cleanup (P5–P9)

- **P5** rolling 20/60-day correlation monitor; decoupling treated as its own signal.
- **P6** reconcile COT against your own framework (stop deferring to EdgeFinder); verify macro snapshots aren't data drift.
- **P7** split geopolitics into priced-in vs fresh-shock.
- **P8** London-sweep invalidation: objective N-bar / X-point reclaim trigger.
- **P9** VWAP exit: anchor confirmed = session VWAP; require close-beyond + **structural confirmation**, not VWAP alone.

Phase 5 — Hygiene (P10–P13, runs in parallel)

- **P10** timestamp + staleness flag on every input.
- **P11** event blackout: flat-before-print windows for NFP/CPI/FOMC (Warsh's first FOMC = Jun 16–17). Frozen calendar rail.
- **P12** net R:R — already baked into the harness from Phase 1.
- **P13** stop calling overnight IV a GVZ "proxy"; treat as related, not equal.

Phase 6 — Rewrite the AI prompt

Once P0–P2 are settled, rewrite the Master Context operating prompt to encode the **fixed** model — orthogonalized scoring, regime conditioning, risk gating, freshness flags — not the current one. (The audit explicitly sequences this last.)

3. Requirements & status

Data

#	Requirement	Status
D1	Gold D1 (17y) — regime + PCA	● Source confirmed; file to download/place
D2	Gold M5 (3y) — session structure	● Source confirmed; file to download/place
D3	Gold M1 (~7mo) — sweep micro-timing	● Source confirmed; file to download/place
D4	XAG, EUR/USD, USD/JPY D1 — factors	● Source confirmed (broker has them)
D5	FRED 10y nominal / 10y real / broad USD	● Automated by build_panel.py
D6	daily_panel.csv generated	● Pending — run build_panel.py
D7	Timezone confirmed = GMT/UTC	● Done
D8	CSV format verified (tab-sep, space-thousands volume)	● Done

Tools / environment

#	Requirement	Status
T1	TradingView + Pine (execution surface)	● Done — LSE v5 exists
T2	Python 3 + pandas, numpy, requests	● Confirm installed on your machine
T3	scikit-learn + (hmmlearn or statsmodels) for P0	● Install at Phase 1

#	Requirement	Status
T4	matplotlib (equity curves / tables)	● Install at Phase 1
T5	Research repo/folder structure	● Create at Phase 1
T6	Notion journal (P4)	● Exists

Decisions / specs

#	Requirement	Status
S1	Architecture = 3 surfaces, frozen/fluid line	● Confirm in §1
S2	No automation in challenge	● Confirmed
S3	Runtime = Python (lab) + Pine (execution)	● Confirmed
S4	Build order (this roadmap)	● Confirm
S5	Regime granularity for P0	● Open — see §5
S6	Sweep invalidation window (N min / X pts) for P8	● Open — settle at Phase 4

Artifacts produced so far

Artifact	Status
<code>build_panel.py</code> (ingest broker CSVs + FRED → panel)	● Written, tested on synthetic; FRED untested live
<code>pull_data.py</code> (Dukascopy fallback)	● Written; superseded for gold, kept as backup
LSE v5 Pine HUD	● Exists (your code)

Legend: ● done · ● awaiting your confirm · ● ready, action on you · ● not started

4. Status summary

- **Done (●):** 8 of 20 requirements — data format/timezone verified, FRED automated, Pine HUD live, Notion exists, runtime + no-automation settled, panel builder written.
- **One action unblocks the most (● D6):** run `build_panel.py` tomorrow → produces `daily_panel.csv`, which unblocks all of Phase 1.
- **Awaiting your sign-off (●):** the architecture (§1) and the build order (§2).
- **Two open decisions (● S5, S6):** only S5 is needed *now*; S6 waits until Phase 4.

Net: the foundation is ~80% in place. The only thing standing between us and the Phase 1 magic moment is: install Python deps, run the panel builder, and answer one design question (§5).

5. The one decision to settle before P0

Regime granularity. How many structural states should the classifier output?

- **2-state — Range vs Trend.** Simplest, most interpretable, easiest to fit reliably on the data we have. Directly inverts the right rules (fade-the-sweep vs ride-the-break).
- **4-state — Trend-up / Trend-down / Range / High-vol-event.** Richer, but needs more data per state to fit stably and risks thin samples in the expectancy table.

My recommendation: start with 2-state. It's robust, it's enough to flip your core rule set, and the volatility axis you already have (ATR-ratio) covers the "high-vol" dimension separately. We can split to 4 later once the expectancy table shows it's warranted — and because the rails are frozen, that upgrade costs nothing downstream.

6. Tomorrow's first moves

1. Confirm §1 (architecture) and §2 (build order).
2. Install: `pip install pandas numpy requests scikit-learn matplotlib`.
3. Run `build_panel.py` → produce `daily_panel.csv`; paste me the printout (shape, columns, NaN counts, tail).
4. Tell me 2-state or 4-state (§5).

Then I spec P0 and we build the regime classifier + the first expectancy table.